<u>Summary for Bayesian Optimization Algorithm</u>

Bayesian optimization algorithm is an algorithm that helps to minimize the evaluation of a given function based on the selection of best next parameters using principal assumptions. In one example of Bayesian optimization algorithm, the function is assumed to be sampled from Gaussian Process and the selection of next hyperparameters is preceded by optimizing either the expected improvement over the current result or the Upper Bound Confidence. The basic problem statement of the algorithm is:

'Find X, while minimizing the number of evaluations of f(x)'.*

Here the function is given as a black-box(where one knows the output of a given function for a given input while not seeing or understanding the underlying expression of the function) and the aim is to approximate this function with less evaluation.The main distinct nature of Bayesian Optimization algorithm is the probabilistic nature of its model. For a simple linear model there might be a large set of hyperparameters where the objective function of the model needs to be evaluated for the purpose of optimization. However, evaluating all parameters at each iteration would be unreasonable. So, the BOA uses the probability decision mechanism to decide which hyperparameters will be part of the next iteration based on current best selection. BOA itself has parameters that must be optimized in order to proceed to the next iteration.These are,Acquisition Function, Kernel(design of the prior), Parameter wrapping and Initializing scheme.

The kernel is first selected to model the random function. The acquisition function then provides knobs for controlling the exploration-exploitation trade-off.Search is proceeded in regions where mean of the kernel is high and exploration is done in regions where the uncertainty, described by variance, is high.

To move on with the algorithm, two choices are made. First the kernel needs to be determined that will supposedly model the black-box function. Also , the acquisition function needs to be defined. With these two decisions the experiment proceeded.

In the first iteration BOA treats the unknown function as a random function placed a prior over it. In the example mentioned above, the Gaussian process was used as prior. Based on this prior the function is evaluated for a finite set of data points and the posterior distribution which will be used as a prior in the next iteration. For successive iterations these processes are performed until the reasonable convergence behavior is observed. These iterations result in finding a minimum of difficult non-convex functions with relatively low evaluations. Despite the reduced number of evaluations for a given function, there is a cost of performing more computations to determine the next point to try.

The main advantage of BOA, as described above , is less evaluation of the black-box function and its main disadvantage is the overhead computation. So unless the requirement constrains the number of evaluation, other methods of optimization will hold. However, the need for less evaluations is required , Bayesian Optimization Algorithm will hold fine.